

How does the roots architecture affect the runoff –infiltration –erosion processes under varying rainfall events?

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ABSTRACT

Plant roots play a critical role in regulating surface–belowground hydrological processes, yet the mechanisms by which root regulates runoff, infiltration and erosion under different rainfall events remain insufficiently understood. In this study, root–soil composites of Moso bamboo (*Phyllostachys edulis*) with three stand densities, including high-density (S1: 2400 plants ha⁻¹), medium-density (S2: 2000 plants ha⁻¹), and low-density (S3: 1600 plants ha⁻¹), were investigated. Controlled rainfall simulations representing low (20 mm h⁻¹), moderate (30 mm h⁻¹), and high (40 mm h⁻¹) rainfall intensities were applied to quantify the dynamic movement processes of surface runoff, subsurface flow, vertical infiltration and surface sediment were collected and quantified using a custom-designed soil water movement monitoring device. Root architecture and soil pore structure were quantified using CT scanning and pore network modelling. Structural equation modelling was further applied to elucidate root–pore–hydrology linkages. Results indicate that (1) under low to moderate rainfall (20–30 mm h⁻¹), subsurface flow was the dominant hydrological component, accounting for 49.87%–88.70% in S1, 50.57%–89.20% in S2, and 41.60%–81.60% in S3 of the total runoff and infiltration amount. In contrast, under high-intensity rainfall (40 mm h⁻¹), surface runoff became dominant, contributing 72.6%–75.0% across all densities, surface runoff rates ranged from 0.052–0.486 L min⁻¹ m⁻² across treatments. (2) Relative to S3, under 20–30 mm h⁻¹, surface runoff and sediment yield in S1 and S2 were 11%–32% and 31%–62% lower, respectively, while subsurface flow in the 10–20 cm and 20–40 cm layers increased by 48.31%–54.68% and 27.76%–30.59%, respectively. However, under 40 mm h⁻¹, no significant density effects were detected for runoff partitioning or sediment yield ($P > 0.05$). (3) Root effects were mainly indirect and mediated by pore structural changes: the total effects of throat radius, connected porosity, and throat length on surface runoff were –0.29, 0.35, and 0.24, on subsurface flow were 0.43, 0.48, and –0.41, and on surface sediment were –0.21, –0.31, and 0.24, respectively. These results suggest the presence of a potential rainfall-intensity threshold, beyond which the regulatory effects of root–pore structure on runoff partitioning and sediment reduction become substantially weaker.

1. Introduction

Rainfall-induced runoff, infiltration, and erosion processes are fundamental components of hillslope hydrology and constitute major drivers of soil degradation in sloping forest ecosystems worldwide (Yuan et al., 2015; Zhang et al., 2019; Utzig et al., 2023). The partitioning of rainfall into surface runoff, subsurface flow, and vertical infiltration directly determines soil water redistribution, sediment transport, and slope stability. In vegetated hillslopes, plant roots strongly regulate

these hydrological processes by altering soil pore geometry, connectivity, and hydraulic conductivity (Yang et al., 2024; Qiu et al., 2024; Zhang et al., 2024). However, despite extensive studies on vegetation effects on erosion control, the mechanistic pathways through which root architecture modifies pore network structure and subsequently controls runoff–infiltration–erosion dynamics under different rainfall conditions remain poorly quantified Table 1.

Runoff generation, soil water infiltration, and sediment transport are tightly coupled processes during rainfall events (Bahddou et al., 2023;

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Table 1
Basic information of sample plots.

Stand density	Slope (°)	Slope aspect	Altitude (m)	Stand age
S1	30	Sunny slope	75	2
S2	29	Sunny slope	72	2
S3	30	Sunny slope	73	2

Fang et al., 2024; Jing et al., 2025). Enhanced infiltration capacity can reduce surface runoff and consequently mitigate soil erosion (Kakeh et al., 2021), whereas preferential flow through connected macropores may accelerate subsurface flow movement, potentially destabilizing soil structure and promoting internal erosion (Mulyono et al., 2018; Balzano et al., 2019). At the same time, surface runoff can alter pore connectivity through sealing and clogging, leading to feedbacks between infiltration and erosion processes (Yang et al., 2020; Wen et al., 2023; Jiao et al., 2024). These nonlinear interactions are strongly controlled by rainfall intensity and duration, making it challenging to disentangle the combined effects of rainfall and root-induced soil structure (Gelete et al., 2023; Guo et al., 2024). Recent advances in ecohydrology have increasingly emphasized the importance of pore-scale processes in controlling rainfall–runoff responses, particularly through CT-based pore network characterization and threshold-based hydrological behavior analyses (Bai et al., 2022; Yang et al., 2022; Chen et al., 2024; Xing et al., 2025; Darshan et al., 2026). These studies have substantially improved the understanding of pore-scale flow dynamics and preferential flow development. Nevertheless, two major limitations remain unresolved. First, most CT-based studies focus primarily on soil pore morphology and hydraulic properties, while the structural regulation imposed by root architecture is rarely incorporated explicitly. Second, vegetation–hydrology studies are commonly conducted at plot or catchment scales, where subsurface pore connectivity and root–pore interactions cannot be directly resolved. Consequently, the structural mechanisms linking root architecture, pore network evolution, and hydrological pathway partitioning under varying rainfall intensities remain unclear. Methodological limitations further hinder progress in this field. Field runoff plot observations provide integrated hydrological responses but offer limited insight into internal subsurface flow redistribution within root–soil composites (Lu et al., 2018; Shi et al., 2022; Sun et al., 2023; Wang et al., 2023; Li et al., 2025b). Meanwhile, many laboratory rainfall simulation studies rely on disturbed soils, which disrupt natural root systems and pore continuity, thereby obscuring the intrinsic coupling between root architecture and preferential flow development. Therefore, experimental approaches that preserve intact root–soil structures while enabling controlled rainfall manipulation and pore-scale characterization are urgently needed to clarify the mechanisms governing hydrological pathway transitions.

Moso bamboo (*Phyllostachys edulis*) dominates subtropical forest ecosystems in southern China and is widely recognized for its effectiveness in soil and water conservation (Li et al., 2019; Li et al., 2024). Its root system is characterized by dense lateral rhizomes and fine roots predominantly distributed within the upper 40 cm of soil, which substantially enhances throat radius and pore connectivity (Liu et al., 2013). Importantly, bamboo root architecture and associated soil pore structures vary with stand density, providing an ideal natural system for examining how root-induced structural heterogeneity influences hydrological and erosion processes under different rainfall regimes. Compared with recent studies that primarily focus either on pore-scale flow simulation or large-scale vegetation–hydrology interactions (Darshan et al., 2026), this study explicitly integrates root architecture, pore network structure, and hydrological responses within a unified experimental framework. By combining undisturbed root–soil composites, rainfall simulation experiments, high-resolution CT scanning, and pore network modeling, this study quantitatively links root-induced pore structural changes with runoff–infiltration–erosion partitioning under different rainfall intensities. More importantly, the study

identifies a rainfall-dependent transition in hydrological regulation, providing new mechanistic insight into how root–pore interactions control hillslope hydrological responses before and after infiltration capacity thresholds are exceeded.

The specific objectives of this study were to: (1) quantify the responses of surface runoff, subsurface flow, vertical infiltration, and soil erosion to varying rainfall intensities across different bamboo stand densities; (2) characterize the vertical and lateral redistribution of soil water under different rainfall intensities; and (3) elucidate the mechanistic role of root-induced pore structure changes in driving spatio-temporal variations in runoff–infiltration–erosion processes.

2. Materials and methods

2.1. Study area

The study area is located in Anji County, Zhejiang Province, with a geographical location of 30°36′–30°39′N, 119°33′–119°36′E (Fig. 1a). This area belongs to a subtropical monsoon climate, with an average annual rainfall of 1400 mm and an average annual temperature of 16.6°C. The soil type is mainly yellow soil, and the thickness of the soil layer is 40–50 cm. Moso bamboo forest is the main vegetation type in this region, accounting for 69.5% of the forest types, with most Moso bamboo aged between 2–3 years (Fig. 1b).

2.2. Plot selection and root–soil composite collection

Taking the different stand density (S1, S2, and S3: 2400, 2000, and 1600 plants ha^{−1}, respectively) of Moso bamboo forests on steep slopes as the research object, within each stand density, three independent sampling plots with similar slope gradient, aspect and soil type were first identified. In each plot, a 100 cm × 100 cm sampling area was delineated, within which a 60 cm × 60 cm core area was randomly selected to minimize edge effects. Soil excavation was performed gradually from the outer boundary toward the core area using hoes, shovels and Luoyang augers. Excavation proceeded to a depth of 50 cm to allow full isolation of the soil column. After shaping and smoothing the surrounding planes, an undisturbed root–soil composite with dimensions of 60 cm × 60 cm × 40 cm (length × width × height) was carefully extracted from the soil surface downward. During extraction, all faces of the soil monolith were trimmed to maintain planar surfaces, and particular care was taken to avoid disturbance, breakage or displacement of roots. To control for any interference from non-target root systems (such as weeds), all surface roots of unwanted vegetation were manually removed before soil excavation. Only the roots of Moso bamboo (*Phyllostachys edulis*) were retained for analysis. Immediately after extraction, each sample was tightly wrapped with 60 cm-wide industrial polyethylene film, ensuring complete coverage of all faces, especially the lateral and bottom surfaces, to prevent deformation and moisture loss during transportation. The original orientation of each sample was clearly marked to preserve the root structure's spatial relationship. A total of nine undisturbed root–soil composite samples (three per stand density) were collected. All samples preserved the in-situ root architecture, soil structure, and pore characteristics. These were then subjected to medical CT scanning for three-dimensional root–pore structure analysis. Soil sampling was conducted using a stratified sampling method, with 10 cm depth intervals from 0 to 40 cm. For each layer, soil samples were collected with three replicates using standard ring samplers and aluminum boxes. These samples were used for determining saturated hydraulic conductivity (Ks), bulk density, and initial gravimetric moisture content. Additionally, approximately 1200 g of loose soil was collected from each layer in sealed plastic bags for the analysis of soil organic carbon, sand, silt, and clay content, and pH. The basic soil properties are shown in Table S1.

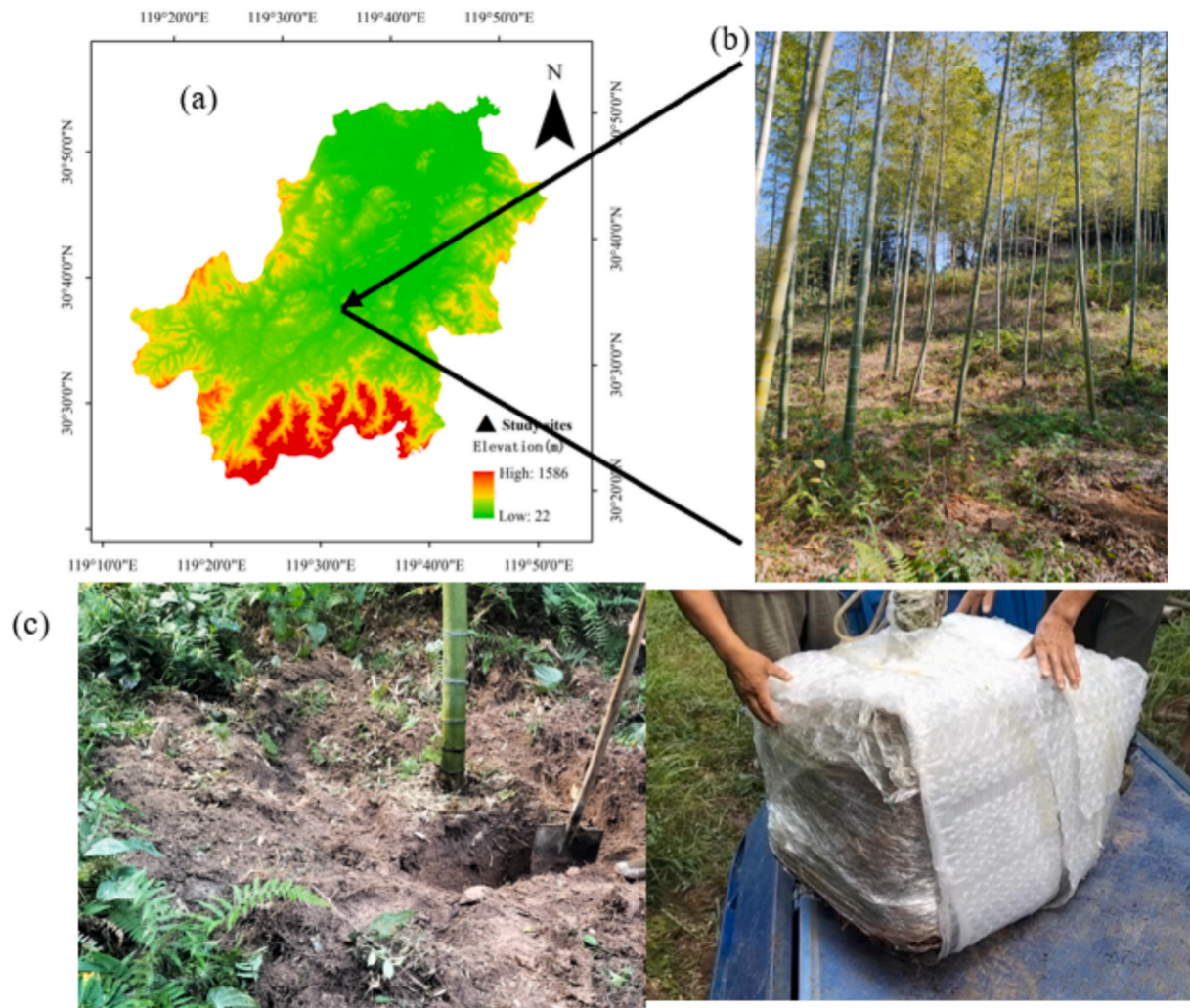


Fig. 1. Location map of the study area (a), Moso bamboo (b) and root-soil composite (c).

2.3. CT scanning and image processing

The root-soil composites with different stand densities were scanned with an Argo NATOM 32 Fit CT scanner. The CT uses the Opti Wave light wave detector with 16 detector rows, achieving the highest visual spatial resolution of 0.64 mm in the field of 32-slice CT, thereby ensuring the clarity and delicacy of CT 3D images. The maximum scanning field of view is 60 cm × 60 cm × 50 cm, and the noise of the scanned image is 0.27%. Considering that medical CT scanners are designed for scanning the human body, the parameters must be adjusted to the scan root-soil composite. After calibration, the peak voltage is set to 130 kV, the current is 200 mA, and the scanning time is 0.8 s per revolution.

Various types of noise in CT images can interfere with the valid information in the image. Accordingly, pretreatment with filtering and noise reduction is required before root architecture and soil pore structure segmentation. To eliminate salt-and-pepper noise introduced during the acquisition and reconstruction processes of CT volume data, a median filtering method was employed for image preprocessing in this study. Specifically, a 3 × 3 × 3 sliding window was used to scan the volume, and the median value of the voxels within each window was calculated to replace the central voxel. This approach effectively suppresses isolated noise while preserving structural edges. Compared to mean filtering, median filtering offers superior edge preservation and is widely used for reducing random noise in volumetric CT datasets (Wang et al., 2019).

Subsequently, root systems and soil macropores were segmented

from CT images using a combination of thresholding and region-based approaches (Gao et al., 2019). All image processing and analyses were conducted using Avizo 9.0 and ImageJ 1.53c. Preprocessed CT slices were converted into 8-bit grayscale images (0–255). Distinct grayscale characteristics allowed differentiation among soil matrix, macropores, and roots. The soil matrix exhibited relatively high grayscale values (>90) with continuous and dense structures, whereas macropores appeared as low grayscale regions (0–66), typically forming connected voids with clear boundaries. Roots were identified as intermediate grayscale structures (30–94), generally exhibiting elongated and branched morphologies with spatial continuity (Melinda et al., 2006; Gao et al., 2019). Macropores were initially segmented using the Otsu thresholding method, which determines the optimal global threshold by maximizing inter-class variance. To minimize misclassification due to grayscale overlap between roots and pores, a logical subtraction operation (“Arithmetic” module in Avizo) was applied to remove regions overlapping with the subsequently extracted root structures. The final pore segmentation results were validated through slice-by-slice comparison with original images and 3D reconstruction. Root extraction was performed using a region-growing algorithm based on grayscale thresholds and spatial continuity. The overall threshold range for root identification was 30–94, with slight variations across soil depths. The extracted root structures were further refined through visual inspection and correction based on continuity, boundary clarity, and 3D morphology. In cases where roots and pores were closely associated or partially overlapping (grayscale range ~ 30–66), segmentation was

refined by integrating grayscale thresholds with morphological continuity and spatial topology criteria. Root characteristic parameters include root length (RL), root surface area (RSA), and root volume (RV).

Pore network modeling was conducted using the Pore Network Extraction (PNE) module in Avizo, which is based on the maximal inscribed sphere algorithm. This approach identifies the largest sphere that can be inscribed within each pore voxel, defines pore bodies as nodes, and determines pore throats through connectivity and clustering procedures (Wardak et al., 2024). The resulting equivalent pore network model was used to quantify structural parameters, including pore throat radius (TR), throat length (TL), and connected porosity (CP). The CT image processing process is shown in Fig. 2.

2.4. Artificial simulated rainfall test

The artificial simulated rainfall test was carried out in the artificial rain hall of Soil and Water Conservation Science and Technology Demonstration Park in Anji County, Zhejiang Province. The rainfall simulators mainly included pressure regulators, flow meters, variable frequency water pump and water supply tank (Fig. 3a). The rainfall height of the device is 3 m, the effective rainfall area is 3 m × 3 m, the rainfall uniformity is about 85%, the rainfall intensity can be adjusted in the range of 10 – 90 mm h⁻¹, and the adjustment accuracy is ± 5 mm h⁻¹. Before each test, the simulation device was repeatedly verified and calibrated by using rain gauges to ensure that the rainfall intensity and rainfall uniformity were consistent with the natural rainfall characteristics, thereby meeting the simulation test requirements of this study.

The soil moisture movement measurement device consists of a runoff box, a deflector tube and water storage buckets (Fig. 3b). The size of the runoff box of this device is 0.6 m × 0.6 m × 0.6 m, and the side and rear plates can be moved to ensure that the water does not leak out of the side wall. To separately quantify different hydrological components, the device was designed with two distinct water collection systems. Sub-surface flow was collected through 3,600 lateral perforations (1 mm diameter) distributed evenly along the side walls, allowing lateral flow within the soil profile to be intercepted. Vertical infiltration was collected through 36 bottom outlets (1 cm diameter), which captured percolating water moving downward through the soil column. Surface runoff was collected at the downslope outlet of the runoff box. A deflector system guided the different flow components into separate collection containers, enabling independent measurement of surface runoff, subsurface flow, and vertical infiltration. In addition, the adjustable slope range of the device is 0–60°, which can simulate and observe the dynamic change process of vertical and horizontal infiltration of soil moisture in undisturbed soil columns with different slopes. To further improve clarity, a cross-sectional schematic diagram with flow pathways has been provided in the [Supplementary Materials](#)

(Fig. S1), illustrating the internal structure and the separation of flow components. This design allows simultaneous and independent quantification of multiple flow pathways, improving the reliability of hydrological partitioning measurements.

The root–soil composites after X–CT scanning was placed directly on a soil moisture movement measurement device. Meanwhile, the side and back plates of the runoff box were adjusted to make full contact with the soil, and the slope of the device was fixed at 30°. The rainfall simulation protocol involved a pre-wetting phase where a light rainfall intensity was applied to each root–soil composite before each rainfall experiment to standardize the initial soil moisture conditions. This pre-wetting continued until surface runoff was observed, signaling that the soil moisture had reached a comparable initial condition across all samples. The pre-wetting phase was carefully controlled to avoid significant soil structure disturbance, as only a low intensity rainfall was applied. After pre-wetting, the samples were allowed to rest for 24 h under covered conditions, ensuring moisture redistribution and partial recovery of pore connectivity.

Controlled rainfall simulations representing low (20 mm h⁻¹), moderate (30 mm h⁻¹), and high (40 mm h⁻¹) intensities were applied. These rainfall intensities were selected based on the typical rainfall characteristics of the study area and previous rainfall simulation studies, representing commonly observed low, moderate, and high rainfall events in subtropical regions. We used the aforementioned three typical rainfall intensities to conduct artificial rainfall experiments on the root–soil composites, aiming to explore the soil moisture migration process of the soil column of on steep slope under continuous rainfall events, with the device slope adjusted to 30°. The rainfall duration was 1 h, with a 24 h interval. The experiment involved 27 rainfall experiments using nine independent root–soil composite samples, with each sample subjected to only one rainfall intensity. Specifically, for each stand density (S1, S2, S3), three separate samples (e.g., S1-1, S1-2, S1-3) were assigned, and each sample was tested exclusively under one rainfall intensity (20, 30, or 40 mm h⁻¹), with three repetitions for each intensity. This ensures that the experiments are independent and avoids any potential bias from using the same soil monolith for multiple intensities. During the artificial rainfall, the initial surface runoff time, subsurface flow and vertical infiltration time were recorded. As the rainfall progressed, the surface runoff, vertical infiltration and subsurface flow of the different soil layers were recorded every 5 min. After standing for 12 h, the runoff sediment sample was poured out of the upper layer of water and then directly placed in an oven at a temperature of 105°C for 8 h to calculate the runoff and sediment yield. The dynamic variation characteristics of the surface runoff rate, subsurface flow rate, vertical infiltration rate and sediment rate were calculated based on the accumulated water volume (Fang et al., 2024; Li et al., 2025a). After the rainfall ended, we continued to recording for 1 h to account for the hysteresis of subsurface

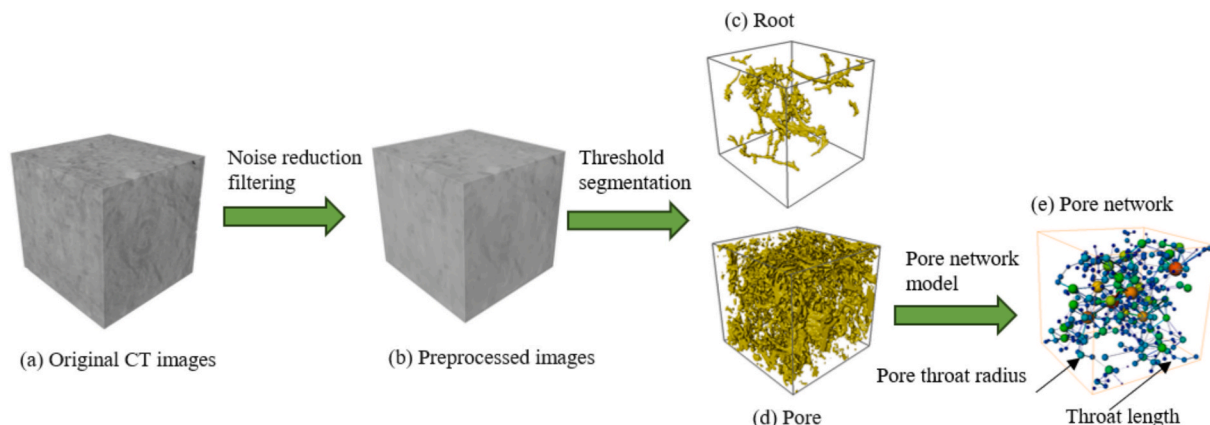


Fig. 2. CT Images Processing Workflow.

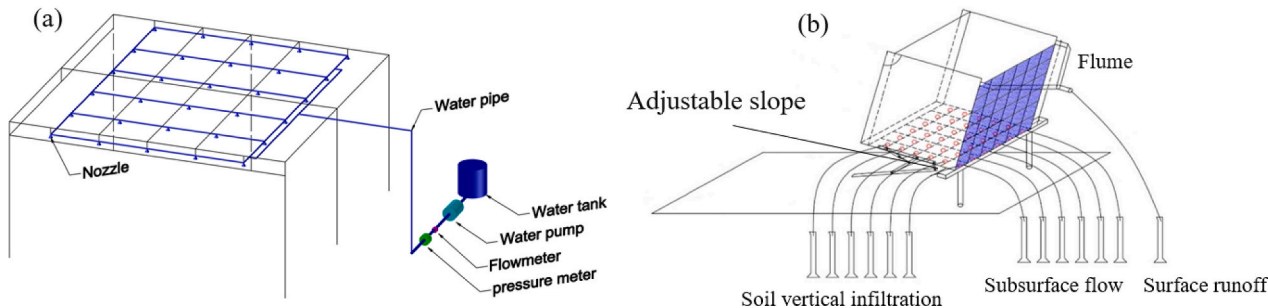


Fig. 3. (a) Artificial rain simulation equipment; (b) Soil moisture movement device.

flow and vertical infiltration.

2.5. Statistical analysis and data visualization

All data processing, statistical analyses, and visualization were performed using SPSS Statistics 27.0.1, Microsoft Excel 2021, Origin Pro 2021, and AMOS 26.0 for structural equation modeling (SEM). A factorial experimental design was employed to evaluate the effects of stand density on root architecture, soil pore structure, and hydrological responses. One-way ANOVA was used to test differences among stand

densities for root traits, pore indices, surface runoff, subsurface flow, and sediment yield, with Duncan’s multiple range test for post-hoc comparisons at $P < 0.05$.

Pearson correlation analysis was conducted to examine relationships among root architecture, pore structure, and hydrological variables. To quantify coupling relationships and potential causal pathways, SEM was employed based on theoretical and empirical evidence, estimating both direct and indirect effects of root and pore variables on surface runoff, subsurface flow, and sediment yield. Model fit was evaluated using χ^2/df , RMSEA, and CFI, with acceptable thresholds of $\chi^2/df = 1-3$, RMSEA

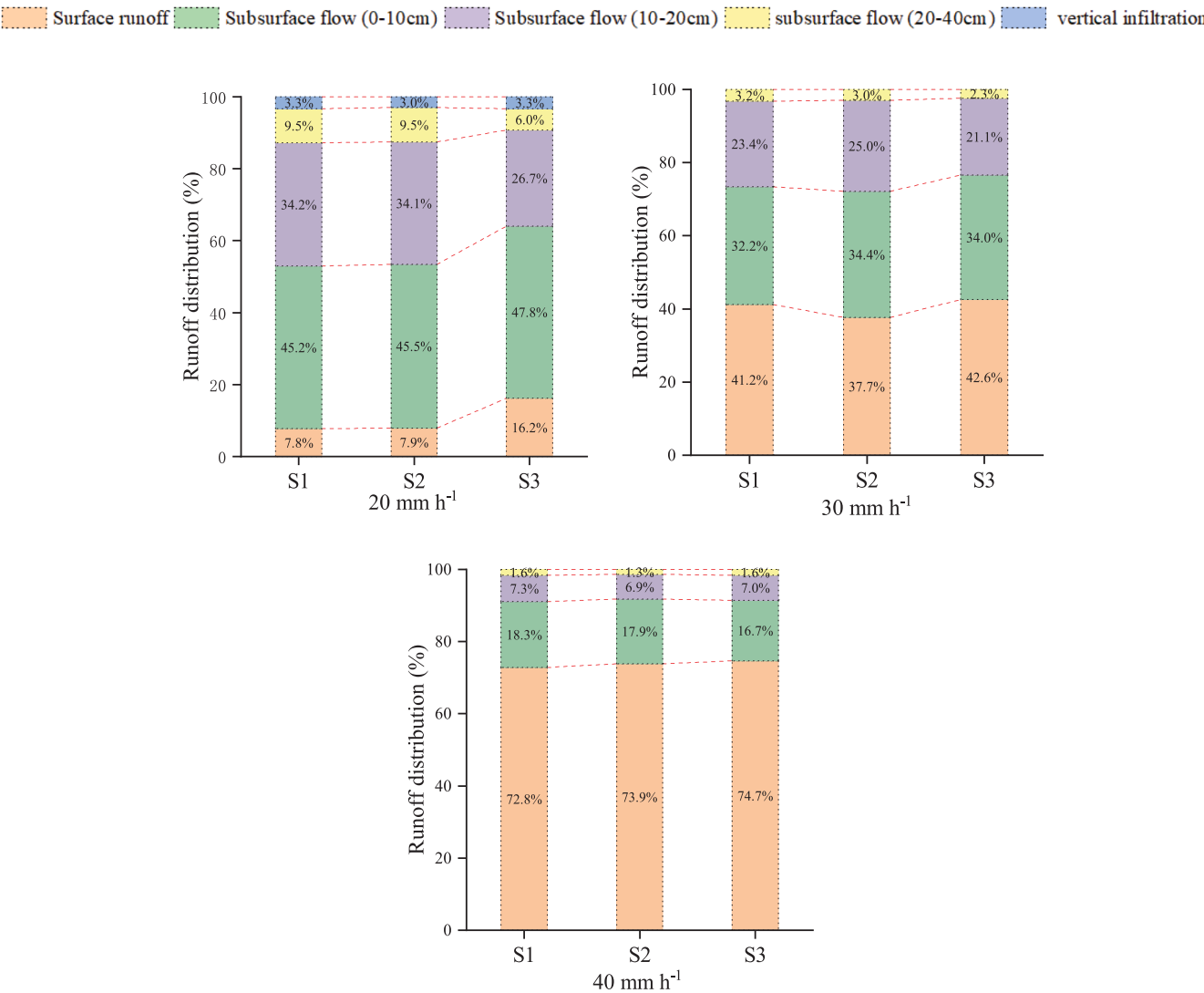


Fig. 4. Runoff distribution under different rainfall intensities and stand densities.

< 0.08, and CFI > 0.90 (Li et al., 2025). Standardized path coefficients and total effects were reported to indicate the relative importance of each variable.

3. Results

3.1. Rainfall-driven runoff partitioning

Rainfall intensity exerted a dominant control on hydrological partitioning among surface runoff, subsurface flow, and vertical infiltration across all root–soil composites (Fig. 3). A consistent transition in dominant flow pathways was observed with increasing rainfall intensity. Under low to moderate rainfall intensities (20–30 mm h⁻¹), subsurface flow was the primary hydrological component, accounting for the majority of total water redistribution, while surface runoff remained relatively limited. In contrast, under high rainfall intensity (40 mm h⁻¹), surface runoff became dominant, exceeding 70% of the total runoff and infiltration amount, whereas subsurface flow was strongly suppressed and vertical infiltration was negligible. This shift indicates a rainfall-driven transition from infiltration-dominated to runoff-dominated hydrological behavior. Across all treatments, subsurface flow was mainly concentrated in shallow soil layers (0–20 cm), while contributions from deeper layers (20–40 cm) declined markedly with increasing rainfall intensity, suggesting reduced downward and lateral redistribution under intense rainfall.

3.2. Surface runoff dynamics

Surface runoff exhibited consistent temporal responses to rainfall intensity (Fig. 4). Under low to moderate rainfall (20–30 mm h⁻¹), runoff rates increased gradually with rainfall duration, reflecting progressive reduction in soil infiltration capacity. In contrast, under high rainfall intensity (40 mm h⁻¹), runoff increased rapidly at the onset of rainfall and quickly approached a quasi-steady state. Stand density significantly influenced surface runoff under low to moderate rainfall intensities ($P < 0.05$; Table 2), with lower-density treatments consistently generating higher runoff rates. However, this effect disappeared under high rainfall intensity ($P > 0.05$), indicating that rainfall intensity overrides root-mediated regulation of runoff generation.

3.3. Soil water infiltration

3.3.1. Subsurface flow

Subsurface flow exhibited clear vertical stratification and strong sensitivity to rainfall intensity (Fig. 5). Across all treatments, subsurface flow was concentrated mainly within the shallow soil layers (0–20 cm), while the contribution of the 20–40 cm layer remained comparatively low. Temporal variations generally followed a unimodal pattern, with flow rates increasing during the initial rainfall stage and gradually declining thereafter.

Table 2

The average surface runoff rate of different root–soil composites under different rainfall intensities.

Rainfall intensity	Stand density	Average surface runoff rates(L min ⁻¹ m ⁻²)
20 mm h ⁻¹	S1	0.052 ± 0.014b
	S2	0.053 ± 0.013b
	S3	0.075 ± 0.017a
30 mm h ⁻¹	S1	0.222 ± 0.025a
	S2	0.227 ± 0.036a
	S3	0.249 ± 0.027a
40 mm h ⁻¹	S1	0.475 ± 0.001a
	S2	0.482 ± 0.002a
	S3	0.486 ± 0.001a

Notes: Different lowercase letters indicate differences in the average surface runoff rate under different densities.

Under low to moderate rainfall intensities (20–30 mm h⁻¹), stand density significantly influenced subsurface flow distribution. High-density root–soil composites (S1 and S2) promoted greater subsurface flow in the 10–20 cm and 20–40 cm layers, indicating enhanced lateral redistribution of infiltrated water through connected pore networks. In contrast, shallow-layer (0–10 cm) flow was relatively greater in S3.

With increasing rainfall intensity (40 mm h⁻¹), subsurface flow decreased markedly across all soil layers, and differences among stand densities became statistically insignificant ($P > 0.05$). Under these conditions, rapid surface runoff generation dominated the hydrological response, while the contribution of subsurface flow to total water redistribution became limited. Overall, the results indicate that increasing rainfall intensity progressively weakened the influence of root-induced pore structures on subsurface water movement.

3.3.2. Soil vertical infiltration

The vertical infiltration rate of root–soil composites with different stand densities under different rainfall events is shown in Fig. 6. The maximum vertical infiltration rates of S1 density, S2 density and S3 density were 0.018 L min⁻¹ m², 0.016 L min⁻¹ m² and 0.02 L min⁻¹ m², respectively. There was no difference in the rate of vertical infiltration in different densities. In addition, when the rainfall intensity was 30 mm h⁻¹, the amount of vertical infiltration was zero.

3.4. Surface sediment

Surface sediment transport responded nonlinearly to rainfall intensity and root–soil composite density (Fig. 7). No detectable sediment transport occurred during the first rainfall event, indicating that initial rainfall primarily contributed to soil wetting. During subsequent rainfall events, sediment transport rates increased markedly with rainfall duration. Under low and moderate rainfall intensities, mean sediment transport rates differed significantly among densities ($P < 0.05$; Table 3), increasing with decreasing stand density. This pattern was consistent with the observed differences in surface runoff rates. In contrast, under high rainfall intensity, sediment transport rates did not differ significantly among densities ($P > 0.05$). Regression analysis revealed a significant quadratic relationship between surface runoff rate and sediment transport rate under low and moderate rainfall intensities ($R^2 = 0.69$ and 0.52 , respectively; Fig. 8). However, this relationship weakened substantially under high rainfall intensity ($R^2 = 0.14$), indicating that sediment transport became increasingly controlled by rainfall kinetic energy rather than runoff hydraulics.

3.5. Root and soil pore structure

The root characteristics of the root–soil composites with different stand densities are shown Table S2. Significant differences can be observed in the root diameter, root length, root surface area and root volume of the root–soil composites of the three stand densities. There was no significant difference in root inclination between S1, S2 and S3. The distribution characteristics of different root inclination is shown in Fig. S3. In the 0–20 cm soil layer, the root inclination angles > 60°, 30–60°, and < 30° accounted for 40.8%–60.2%, 37.6%–39.5%, and 2%–19.7% of the total root volume amount, respectively. In the 20–40 cm soil layer, the root volume with root inclination angles > 60°, 30–60° and < 30° accounted for 5.9%–39.1%, 32.7%–64.1% and 25.8%–29.9%, respectively. In summary, the root inclination angle of 60°–90° in the root–soil composite of *Moso bamboo* was the largest, and the root system growing in the horizontal direction was the most, and it was the most obvious in the 0–20 cm soil layer.

The characteristic parameters of the soil pore structure of root–soil composites with different stand densities are shown Table S3. Moreover, significant differences are observed in Euler number, connected porosity, throat radius and throat length of the root–soil composites of the three stand densities.

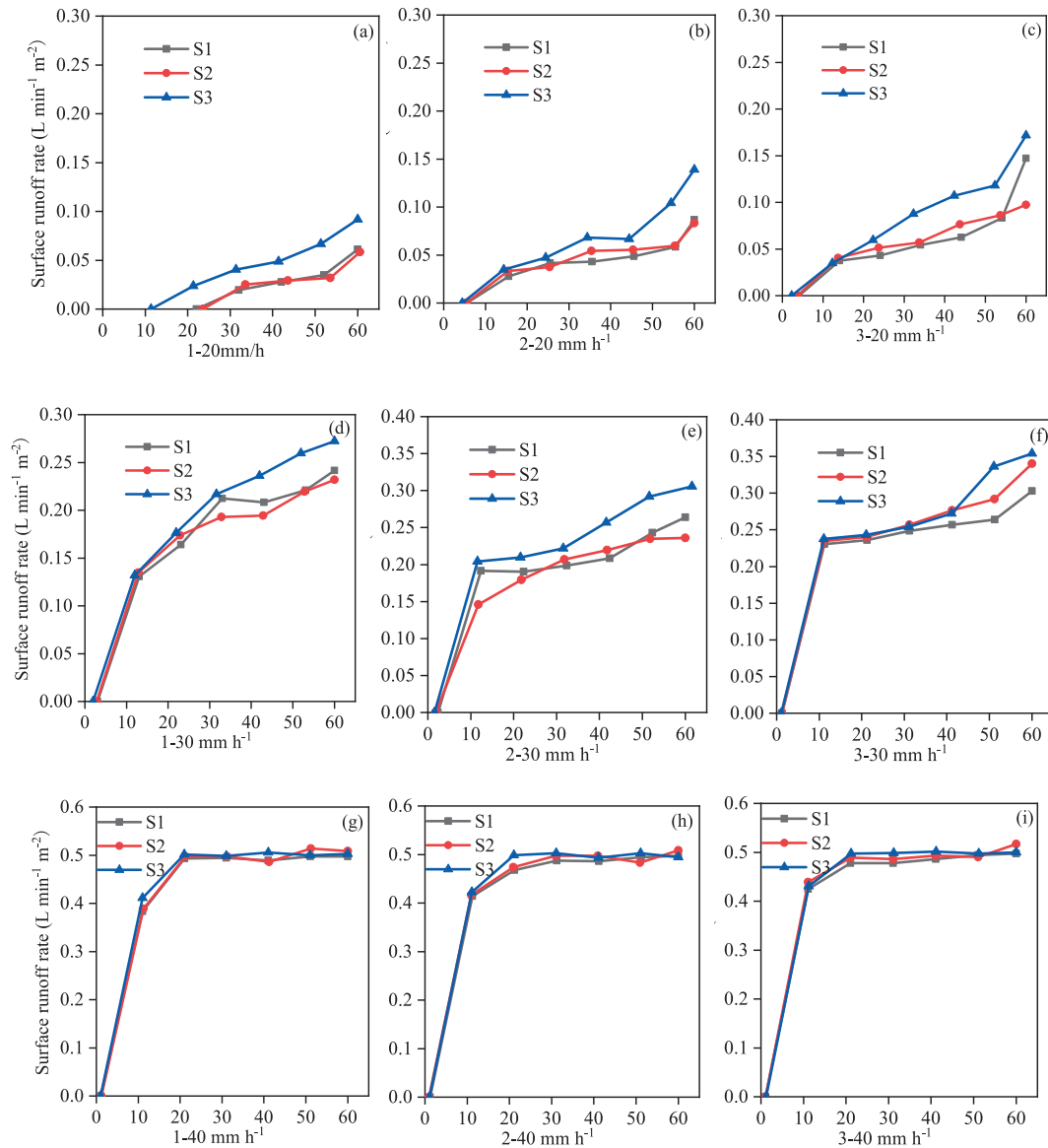


Fig. 5. Surface runoff rate with different rainfall intensities (a–c: 20 mm h⁻¹, d–f: 30 mm h⁻¹, g–i: 40 mm h⁻¹).

The correlation analysis between root architecture and soil pore characteristics is presented in Table S4. Root distribution patterns significantly influenced pore network development and connectivity. Root surface area was positively correlated with throat radius, connected porosity, and pore throat radius, indicating that denser fine-root systems promoted the formation of larger and more connected pore channels. In contrast, root surface area was negatively correlated with Euler number and throat length, suggesting that increased root proliferation enhanced pore network connectivity while reducing pore fragmentation and tortuous flow paths.

Similarly, root volume showed a significant positive correlation with connected porosity and a negative correlation with throat length, implying that larger root systems facilitated the development of continuous macropore pathways favorable for lateral water redistribution. Root length was positively associated with connected porosity and pore throat radius, further demonstrating that elongated root networks contributed to improved pore continuity and flow efficiency within the soil profile. These structural changes are likely to enhance preferential flow development and subsurface flow transport under rainfall conditions.

However, root inclination exhibited no significant correlation with

pore structural parameters, suggesting that pore network development was more strongly controlled by root abundance and spatial proliferation than by root orientation alone. Overall, the results indicate that root-induced modifications of pore geometry and connectivity play an important role in regulating soil water movement and preferential flow pathways.

3.6. Influence mechanism of runoff–infiltration–erosion processes

Structural equation modeling (SEM) was applied to quantify the pathways linking root architecture, soil pore structure, and runoff–infiltration–erosion processes under different rainfall intensities (Fig. 9). In the model, root traits (root length and root surface area) were specified as exogenous variables, pore structural parameters (connected porosity, pore throat radius, and throat length) as mediating variables, and hydrological responses (surface runoff, subsurface flow, and surface sediment) as endogenous variables. Both direct and indirect effects were estimated, and total effects were calculated as the sum of direct and mediated effects. Model performance was evaluated using multiple goodness-of-fit indices (χ^2/df , RMSEA, and CFI), and all models satisfied commonly accepted thresholds ($\chi^2/\text{df} = 2.23$, 1.98, RMSEA = 0.06,

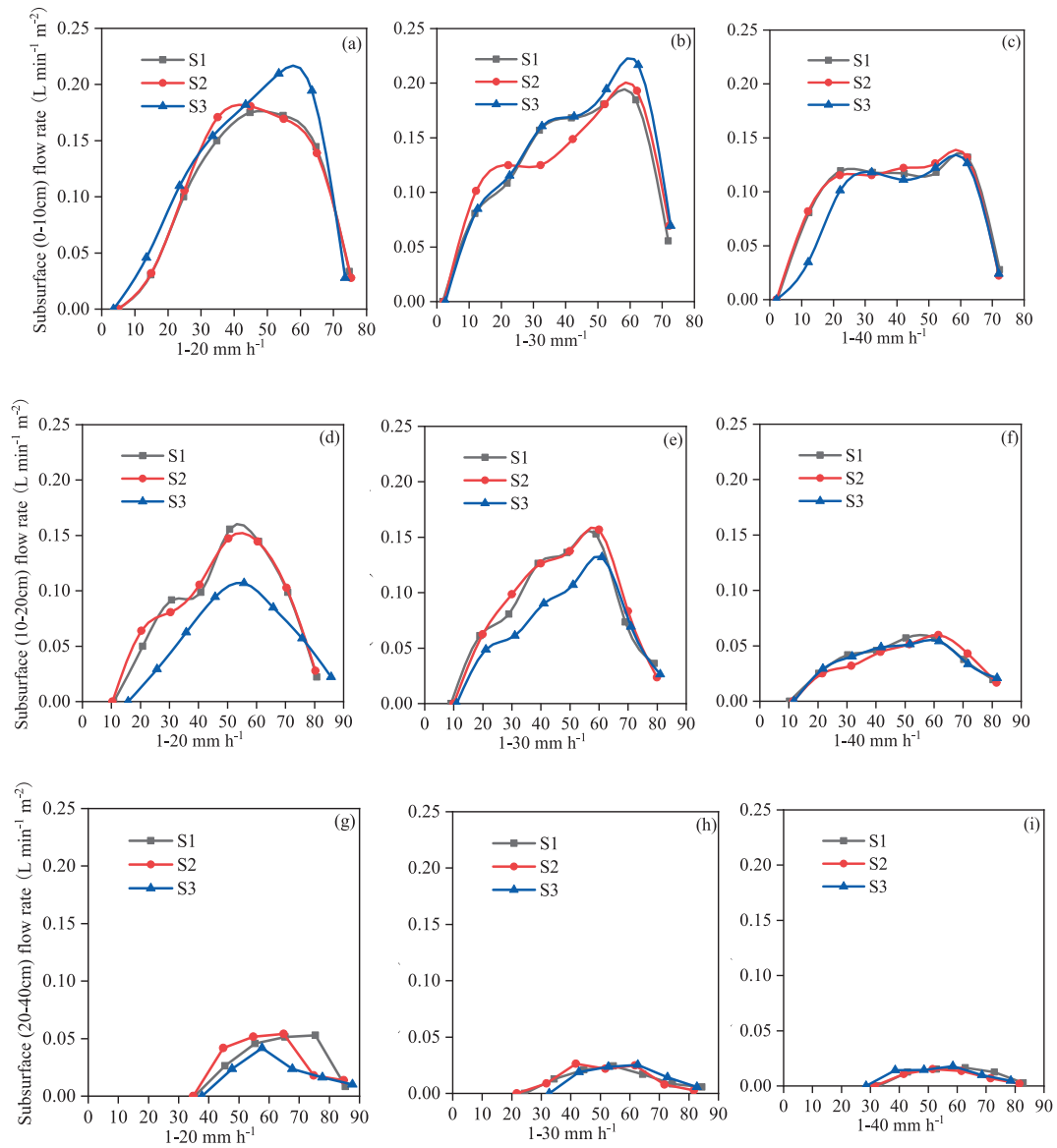


Fig. 6. Subsurface flow rate in different soil layers (a–c: 0–10 cm, d–f: 10–20 cm, g–i: 20–40 cm) under different rainfall events.

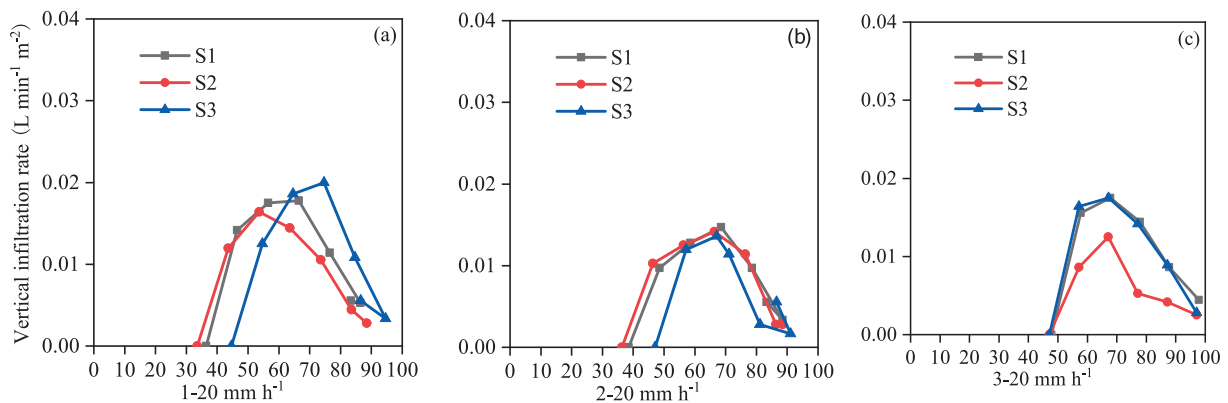


Fig. 7. Soil vertical infiltration rate under different rainfall events.

0.04, CFI = 0.93, 0.96), indicating a good fit. The explanatory power of the model was further assessed using R^2 values for endogenous variables Fig. 10.

Across rainfall intensities, root effects on hydrological and erosion responses were primarily indirect and mediated through pore structural modifications. Root length and root surface area influenced hydrological

Table 3

The average surface sediment rate of different root–soil composites under different rainfall intensities.

Rainfall intensity	Stand density	Surface sediment rate ($\text{g min}^{-1}\text{m}^{-2}$)
20 mm h^{-1}	S1	$0.093 \pm 0.080\text{a}$
	S2	$0.155 \pm 0.029\text{b}$
	S3	$0.244 \pm 0.037\text{c}$
30 mm h^{-1}	S1	$0.326 \pm 0.025\text{a}$
	S2	$0.345 \pm 0.129\text{a}$
	S3	$0.466 \pm 0.143\text{b}$
40 mm h^{-1}	S1	$0.561 \pm 0.051\text{a}$
	S2	$0.595 \pm 0.035\text{a}$
	S3	$0.622 \pm 0.021\text{a}$

partitioning mainly by altering connected porosity, pore throat radius, and throat length, which jointly regulated water redistribution

pathways within the soil profile. Larger connected porosity and pore throat radius enhanced lateral subsurface flow, whereas longer throat length reduced flow connectivity and promoted surface runoff and sediment export. Under low to moderate rainfall intensities (20–30 mm h^{-1}), these root-mediated pore effects were pronounced. For example, the total effects of pore throat radius, connected porosity, and throat length on surface runoff were -0.29 , 0.35 , and 0.24 , respectively, while their effects on subsurface flow were 0.43 , 0.48 , and -0.41 . In contrast, with increasing rainfall intensity, the strength of these relationships declined. Under high rainfall intensity (40 mm h^{-1}), no significant pathways between root–pore characteristics and hydrological responses were detected ($P > 0.05$; Table S5), indicating that rainfall intensity overrides structural controls. Overall, the SEM results suggest a rainfall-dependent threshold in root–pore regulation, whereby structural controls are effective under low to moderate rainfall but become negligible

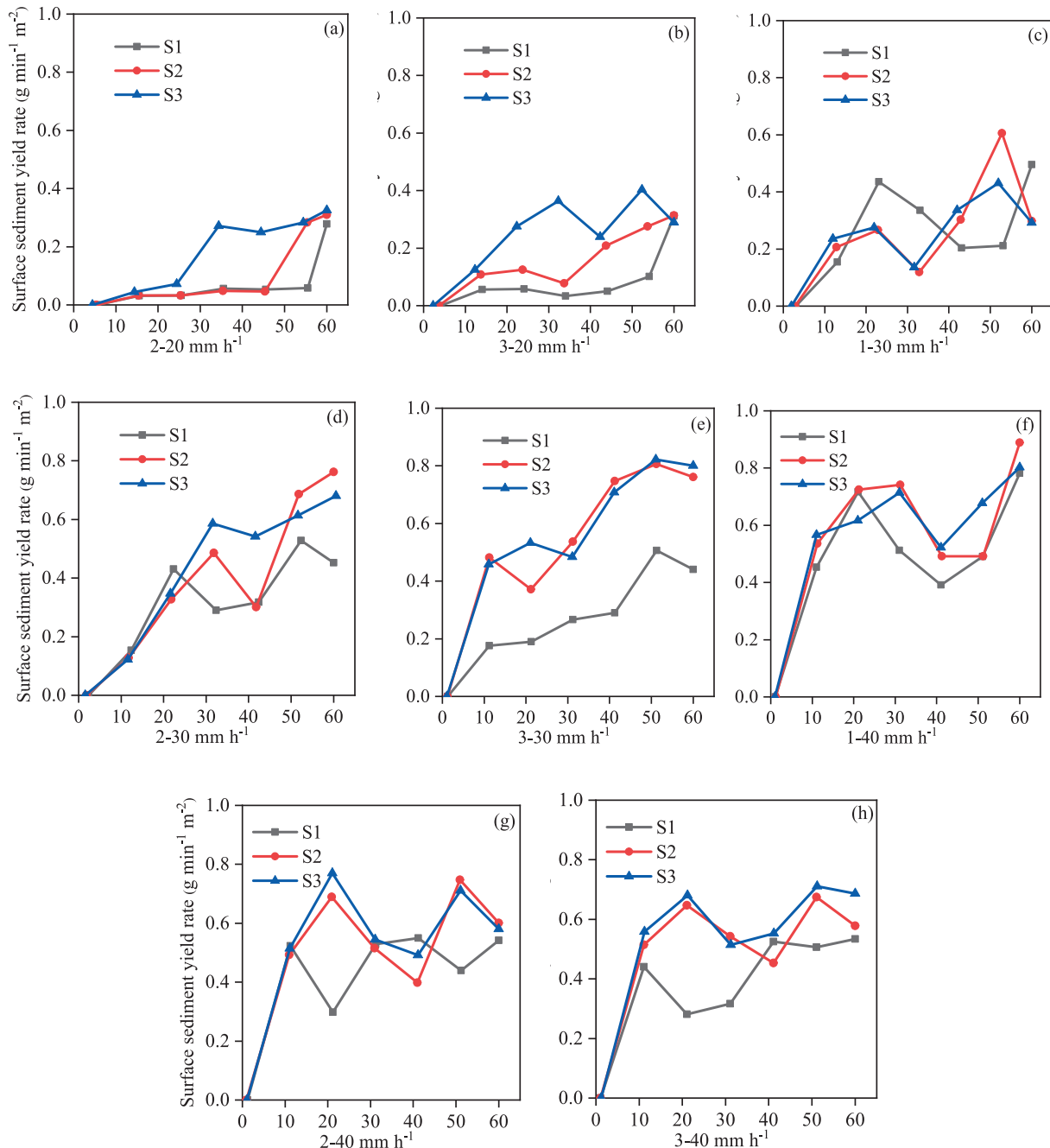


Fig. 8. Surface sediment rate under different rainfall intensities (a–b: 20 mm h^{-1} , c–e: 30 mm h^{-1} , f–h: 40 mm h^{-1}).

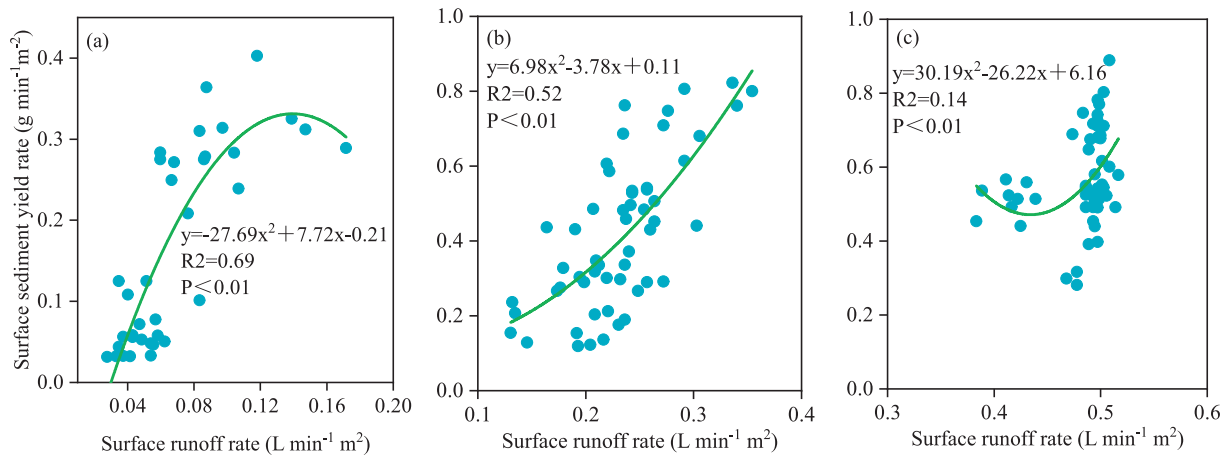


Fig. 9. The relationship between surface runoff rate and sediment rate under different rainfall intensities rainfall intensities (a: 20 mm h⁻¹, b: 30 mm h⁻¹, c: 40 mm h⁻¹).

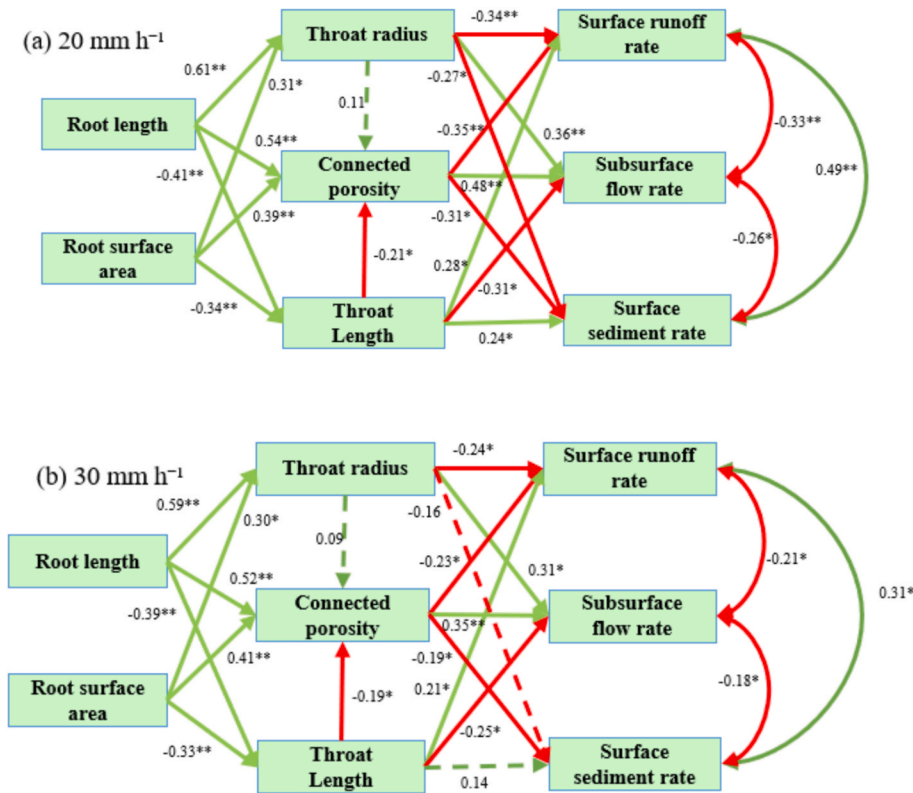


Fig. 10. Direct and indirect effects of root and soil pore structure parameters on surface runoff, subsurface flow and surface sediment rate under varying rainfall events.

once rainfall intensity exceeds the soil's infiltration capacity Fig. 11.

4. Discussion

4.1. Runoff distribution of different rainfall events

A growing body of research demonstrated that rainfall intensity and root architecture are key factors governing runoff distribution (Cai et al., 2024; Han et al., 2024). This study demonstrates that the partitioning of rainfall into surface runoff, subsurface flow, and vertical infiltration is strongly regulated by the interaction between rainfall intensity and root-soil structural characteristics. Under low to moderate rainfall intensities (20–30 mm h⁻¹), subsurface flow dominated the hydrological

response of all Moso bamboo root-soil composites, whereas surface runoff became the prevailing component under high rainfall intensity (40 mm h⁻¹). This transition reflects a threshold-like response of the root-pore system to increasing rainfall forcing. At 20 and 30 mm h⁻¹, the extensive lateral distribution of Moso bamboo roots in the upper soil layers (0–20 cm) substantially enhanced macroporosity and lateral pore connectivity, promoting rapid redistribution of infiltrated water as subsurface flow. Roots act not only as physical conduits but also as architects of connected pore networks, facilitating lateral water movement along root-soil interfaces (Xu et al., 2025). Similar mechanisms have been reported in forested hillslopes, where horizontally oriented roots preferentially enhance lateral flow pathways and delay surface runoff generation (Li et al., 2023; Xu et al., 2025). In contrast, when rainfall

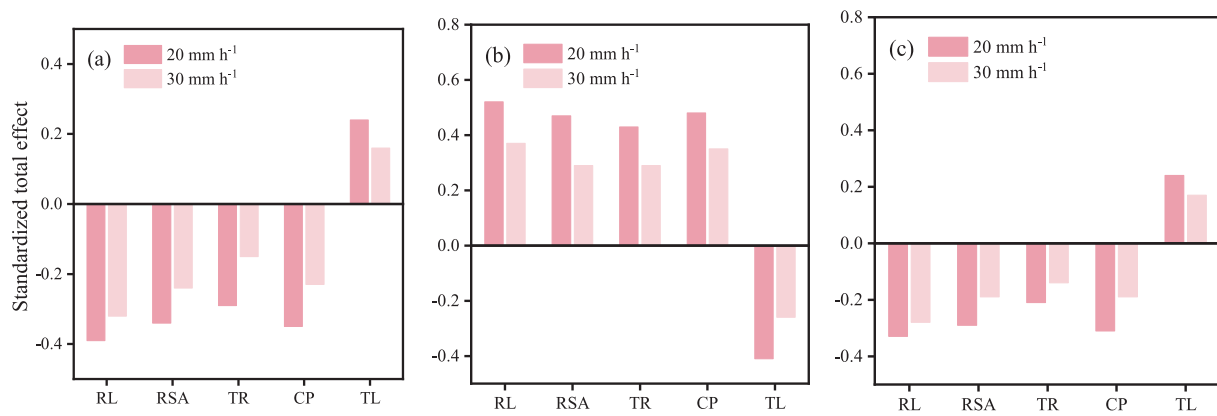


Fig. 11. Standardized total effects of root and soil pore structure on surface runoff (a) and subsurface flow (b), surface sediment (c).

intensity increased to 40 mm h⁻¹, the infiltration capacity of the soil–root system was progressively exceeded. Raindrop impact and surface sealing reduced effective pore connectivity near the soil surface, suppressing subsurface flow and triggering a sharp increase in surface runoff (Carmi et al., 2018). This finding indicates that root-enhanced infiltration is effective only within a finite rainfall range and can be overwhelmed under extreme rainfall conditions (Han et al., 2024).

4.2. Root–pore structural controls on runoff–infiltration–erosion coupling

The differences in runoff, subsurface flow, and sediment transport among stand densities can be attributed to variations in root architecture and associated pore network properties. Root length and root surface area emerged as the dominant biological drivers influencing hydrological and erosional processes, primarily by modifying pore throat radius, connected porosity, and throat length. These pore-scale parameters collectively determine both the capacity of subsurface water transmission. At low to moderate rainfall intensities, higher stand densities (S1 and S2) exhibited lower surface runoff and sediment yield but higher subsurface flow in deeper soil layers (10–40 cm) compared to low-density stands (S3). This vertically differentiated pore network structure reflects that while S3's upper layer had larger pores and higher connected porosity, S1 and S2 maintained more continuous and efficient pore pathways in deeper layers. As a result, water infiltrating through the upper layer in S1 and S2 was redistributed more effectively downward and laterally, which reduced surface runoff and erosion. Structural equation modeling further shows that roots indirectly regulate runoff and erosion by enhancing pore geometry and connectivity, promoting subsurface flow and reducing surface runoff and sediment detachment. These results highlight the importance of subsurface hydraulic architecture, shaped by roots, in erosion mitigation on vegetated slopes (Lin et al., 2019; Zeng et al., 2025).

4.3. Rainfall intensity thresholds and the diminishing role of root–pore regulation

A key finding of this study is the identification of a rainfall intensity range beyond which the influence of root–pore structural characteristics on hydrological and erosional processes becomes less pronounced. At 40 mm h⁻¹, differences among stand densities in surface runoff, subsurface flow, and sediment yield were not statistically significant, despite clear contrasts in root architecture and pore structure. This pattern suggests that under high-intensity rainfall, physical processes such as rapid soil saturation and surface sealing may play a dominant role, thereby reducing the observable influence of biological factors. Under such conditions, infiltration rates are likely to approach their upper limits, and excess rainfall is rapidly converted into surface runoff, which may weaken the linkage between subsurface structure and

hydrological response (Liu et al., 2011; Wei et al., 2024). Similar patterns have been reported in previous studies, indicating that vegetation effects on infiltration and erosion tend to diminish under high rainfall intensity or prolonged rainfall duration (Han et al., 2024). It should be noted, however, that these interpretations are based on observed associations rather than direct causal inference. The SEM analysis provides a useful framework for identifying potential pathways linking root traits, pore structure, and hydrological responses, but does not establish definitive causality. Overall, the results suggest that vegetation-based soil and water conservation measures may be more effective under low to moderate rainfall conditions, while their relative influence may decrease under extreme rainfall events.

4.4. Implications and limitations

This study suggests that root architecture and pore network structure are associated with runoff–infiltration–erosion dynamics, with root-induced pore connectivity potentially influencing slope hydrological responses under the investigated conditions. However, several limitations should be noted. First, although the intact root–soil composite experiments allow for controlled investigation of root–pore mechanisms, caution is needed when extrapolating these results to field-scale slopes with heterogeneous root distributions, variable macropore continuity, and complex boundary conditions. Second, medical CT scanning used in this study only resolves large pores (>0.6 mm); higher-resolution imaging (e.g., industrial CT) is needed to capture multi-scale pore and root structures more accurately. Finally, environmental factors such as temperature fluctuations affecting evapotranspiration, and surface crusting during rainfall, may influence infiltration measurements and should be considered in future experiments. These considerations highlight opportunities for methodological refinement and improved parameterization in scaling root–pore–hydrology interactions from controlled experiments to natural hillslope systems.

5. Conclusion

This study provides new insights into how root-induced pore structure governs hydrological partitioning and erosion processes under varying rainfall intensities. The results reveal a clear transition in dominant flow pathways, shifting from subsurface flow to surface runoff as rainfall intensity increases. Under low to moderate rainfall conditions (20–30 mm h⁻¹), subsurface flow accounted for approximately 41.6%–89.2% of total water redistribution, highlighting the strong control of root–pore connectivity on lateral water movement. In these conditions, higher stand densities reduced surface runoff and sediment yield by 11%–32% and 31%–62%, respectively, while significantly enhancing subsurface flow in deeper soil layers (by up to 54.68%). In contrast, under high rainfall intensity (40 mm h⁻¹), surface runoff dominated

(72.6%–75.0%), and no significant differences among stand densities were observed ($P > 0.05$). Mechanistically, the observed effects of roots on hydrological and erosion responses were closely associated with rainfall-induced variations in pore network structure. Structural equation modeling indicated that pore throat radius, connected porosity, and throat length were important factors influencing runoff partitioning and subsurface flow redistribution. Under low to moderate rainfall intensities, greater pore connectivity was associated with enhanced subsurface flow and reduced surface runoff and sediment transport. However, as rainfall intensity increased, these relationships became weaker, and surface runoff gradually became the dominant hydrological response.

CRediT authorship contribution statement

Lin Zhang: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yutong Li:** Methodology, Investigation, Formal analysis. **Bingchen Wu:** Writing – review & editing, Funding acquisition, Formal analysis. **Shi Qi:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2026.135802>.

Data availability

The data that has been used is confidential.

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